LESSON PLAN

BOMBS & COMPONENTS (32)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	02
Total No. of Lesson Plans	9	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Modification carried out on 7.62mm LMG to convert in to AV gun.
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Modification carried out on 7.62mm LMG to convert in to AV gun.
5. Specific Objectives	32.1.1.1 Trainee will be able to explain requirement of air version guns.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of modification diagrams; Safety: Do not disassemble live weapons; Instructor supervision: Monitor diagram interpretation.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What are the primary operational requirements for air version guns?
(b) How does the environment of an aircraft differ from ground combat?
(c) Why is the 7.62mm LMG selected as the base for the AV gun?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Structural reinforcement for vibration • Feed mechanism modification for reliability • Mounting bracket integration • Cooling system enhancements • Weight reduction measures	L & Ds	PPT, IB	15 min
(b)	• Comparison of ground vs air feed systems • Analysis of vibration dampening techniques • Review of mounting bracket specifications • Discussion on barrel cooling methods • Evaluation of weight reduction strategies	L & Ds	PPT, PDF	20 min
(c)	• Summary of key modification components • Importance of reliability in aerial combat • Role of maintenance in AV gun performance • Future trends in aircraft gun systems • Impact of modifications on aircraft balance	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the specific modifications made to the 7.62mm LMG for AV gun conversion.
(b) Describe how vibration affects the feed mechanism and how it is mitigated.

12. Home Assignment and Follow Up	Prepare a brief note on the differences between ground and air version guns based on the modifications discussed in this lesson.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Introduction to gun alignment occasion to carry on.	Refer to the master syllabus reference list	LP 03

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min